
EXERCISE

Lecture 4: Sealed Bid Auctions

AI508 — Algorithmic Game Theory

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1 Auctions

A typical auction is an “English auctions”

Today we will talk about “Sealed bid auctions”

Definition: Sealed Bid Auctions

An auction with 1 item, n bidders

Each individual bidder i has a private valuation v_i of the item.

At the time of bidding, each individual can at the same time bid on the item. After bidding a winner is chosen.

When designing a SBA, the following things should be considered

1. How should the design infer the amount the winning bidder pays
2. Simplicity of participation (strategic simplicity)
3. Computationally easy to run
4. $(b_1, b_2, \dots, b_n) \rightarrow [n]$ where b_i = bid by bidder i

1.1 First Price Auction

The bidder with the highest bid gets the item by paying what they bid

$$u_i : (b_1, b_2, \dots, b_n) \rightarrow \mathbb{R}$$

$$u_i = \begin{cases} v_i - b_i & \text{if } b_i > b_j \quad \forall j \in [n] \setminus \{i\} \\ 0 & \text{otherwise} \end{cases}$$

1.2 Dominant strategies

Let s be the strategy profile (set of strategies (e.g. bid) by each player)

$$s = \{s_1, s_2, \dots, s_n\}$$

Where s_i be denotes the strategy of player i and $s_{-i} = \{s_1, \dots, s_{i-1}, s_{i+1}, \dots, s_n\}$ (all players besides i)

$$s_i \in S_i; s_{-i} \in S_{-i} = \{S_1 \times S_2 \times S_{i-1} \times S_{i+1} \times S_n\}$$

$$u_i = S_1 \times S_2 \times S_m \rightarrow \mathbb{R}$$

A strategy s_i^* is a dominant (weakly) strategy for player i if:

$$u_i(s_i^*, s_{-i}) \geq \underbrace{u_i(s_i, s_{-i})}_{\text{not } <} \quad \forall s_i \in S_i, \forall s_{-i} \in S_{-i}$$

1.3 Second Price Auction

Highest bidder gets the item, but pays the amount of equal to the second highest bidder

$$B_i = \max_{j \neq i} b_j$$

B would be the max bid, then utility is altered such that

$$u_i = \begin{cases} v_i - B_i & \text{if } v_i > B_i \\ 0 & \text{if } v_i < B_i \end{cases}$$

Then

$$u_i(v_i, b_{-i}) \geq u_i(b_i, b_{-i})$$

v_i is a weakly dominant strategy for player i .